Anurag Singh, M. Sc.

Email | GitHub| LinkedIn | +49 17671724834 | Erlangen 91054

Experiences

Research Asst., FZ-Jülich (IET-2) – Erlangen, Germany *Feb' 2025 – Sept' 2025*

- Python and Excel tool development for end users|PEM-FC and -WE testing
- Electrochemical Experiments and Diagnostics | >10 Cells assembled, tested, and results quantified
- Data analysis | 1-D Modelling | Equivalent Circuit Model fit | SciPy, Pandas, ImpedancePy

Intern/Working Student

Large Systems Development | Sunfire GmbH – Dresden, Germany *Nov' 2023 – Nov' 2024*

- Understanding legacy VLE model and identifying optimisation and error reduction potential
- Parameter hunting based on empirical results | Metaheuristic optimisation
- Prognostication with less than 7% error
- Master Thesis - Using a statistical approach (LBM) to solve NSE using OpenLB (C++ Library)

Project Engineer, Ultratech Cement Ltd. – Mumbai, India *Jan' 2021 – Sep' 2022*

- Interface for asset transformation (across >100 MW WHRS projects)
- Project co-ordination for the head office from FEED to Commissioning
- Feasibility reports | Vendor Evaluations |Presentations |Boilers, Turbines, and Ducting
- Tendering | Documentation | Budget monitoring and control | SAP-MM

Education

Friedrich-Alexander University – M. Sc. Clean Energy Processes *March 2025*

Specialisations - Energy systems	Final Grade - 1,5
Quantitative Methods in Energy Market Modelling	Fuel Cell and Electrolysers (FC & WE)
Economics of Climate Change	Deutsch B2

Master Thesis: Multicomponent Multiphase CFD Simulation (OOP Development)

BITS Pilani – B.E.(Hons.) Mechanical Engineering *July 2020*

Electives: Computer Programming and Engineering Optimisation (MILP, KKT, & constrained calculus)

Miscellaneous

Software: Python, C++, GAMS, SAP MM
Technologies: PEM conversion technologies, LFP Batteries and it BMS, Turbomachinery and Heat Exchangers
Projects: Analysis of hourly Spot Market Price|BESS Dispatch algorithm|Price Forecasting (SARIMAX)
Know-how: Quantitative Analysis: Time Series, P90 Analysis|BESS Revenue Modelling|Redispatch 2.0
Hobbys: Podcasts (*Redefining Energy*, *Pexapark*), Non-Fiction (*e.g. C. Freeman and F. Louçã, D. Yergin*)
Languages: English (C1), German (B2)

Erlangen, 3. March 2026